Web scrapping in Python with BeautifulSoup by Example

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Chapter 1

Introduction

1.1 Installing libraries

Install requests.

```
[autogobble]
    > pip install requests
```

1.2 Introduction

- We will be using the website: `http://quotes.toscrape.com/` which is a website used specifically to webscrap.
- It consists of quotes and tags which you can try to scape.

1.2.1 Webscrapping

- First lets import the required libraries:

```python
import requests
from bs4 import BeautifulSoup
```

- The requests object will provide all the information that the website responded. HTML, status code, etcetera.

```python
website = requests.get('http://quotes.toscrape.com/')
print(website)
# <Response [200]>
```

- The .text attribute in the website gives us the html response. We can use this to send the html string to parse to beautifyl soup. The html string from the site and the type of parser we are going to use must be speficied in the BeautifulSoup object constructor.

```python
website = requests.get('http://quotes.toscrape.com/')
soup = BeautifulSoup(website.text, 'html.parser')
```

- Now we can start to extract elements form this web string.

```python
title = soup.find('title')
print(title)
# <title>Quotes to Scrape</title>
print(title.text)
# Quotes to Scrape
```

- To get the first link on the website:

```python
print(soup.find('a'))
# <a href="/" style="text-decoration: none">Quotes to Scrape</a>
```

- Finding the first quote element using the class attribute:

```python
quote = soup.find(class_='text')
print(quote)
# <span class="text" itemprop="text">The world as we have created it is a process
↪  of our thinking. It cannot be changed without changing our thinking.</span>
```

- If we want all the quote elements we cant use the .find method. Instead we must use the .find_all method.

```python
quotes = soup.find_all(class_='text')
print(quotes)
```

- If we only want the text we can do a for loop:

```python
for q in quotes:
    print(q.text)
```

1.3 Extracting elements using CSS selectors

- Using the previous example, we want to select the css class selector, let's get the first css selector of type text:

```python
quote_text = soup.select_one('.text')
```

- If you want all css selectors that match the .text we can use the .select:

```python
quote = soup.select('.text')
```

1.4 Scrapping the first page

Given the following html in the page provided above:

```html
<div class="quote" itemscope="" itemtype="http://schema.org/CreativeWork">
<span class="text" itemprop="text">The world as we have created it is a process of our
↪  thinking. It cannot be changed without changing our thinking.</span>
<span>by <small class="author" itemprop="author">Albert Einstein</small>
<a href="/author/Albert-Einstein">(about)</a>
</span>
<div class="tags">
```

```
    Tags:
    <meta class="keywords" itemprop="keywords" content="change,deep-thoughts,thinking,world">
    <a class="tag" href="/tag/change/page/1/">change</a>
    <a class="tag" href="/tag/deep-thoughts/page/1/">deep-thoughts</a>
    <a class="tag" href="/tag/thinking/page/1/">thinking</a>
    <a class="tag" href="/tag/world/page/1/">world</a>
</div>
</div>
```

The page has a format, and the format is in divs that contain the quotes. Thus to select all the quote information we can do the following:

```
import requests
from bs4 import BeautifulSoup

website = requests.get('http://quotes.toscrape.com/')
soup = BeautifulSoup(website.text, 'html.parser')
quotes = soup.select('.quote') # select all the class="quotes"
for quote in quotes:
    text = quote.select_one('.text')
    author = quote.select_one('.author')
    tags = quote.select('.tag')
    print(text.text)
    print(author.text)
    for tag in tags:
        print(tag.text)
    print('======')

# The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking.
# Albert Einstein
# change
# deep-thoughts
# thinking
# world
# ======
# It is our choices, Harry, that show what we truly are, far more than our abilities.
# J.K. Rowling
# abilities
# choices
# ======
# There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle.
# Albert Einstein
# inspirational
# life
# live
# miracle
# miracles
# ======
```